Exercise 9 - Frequency Response

1 Introduction

As we have seen in previous exercises, our main goal is to analyze the behavior of dynamical systems. In addition, we aim to design the closed-loop system properties by exploiting the characteristics of the open-loop system, thus avoiding complex analytical computations.
To achieve this, three main graphical methods are commonly used:

- Root Locus (already seen in Exercise 7)
- Bode Plot
- Nyquist Plot

Today and in the coming weeks, we will focus on the Bode Plot, one of the most powerful tools for analyzing the frequency response of linear time-invariant (LTI) systems. However, before studying the Bode Plot, we first need to understand what the frequency response of a system represents.
The frequency response describes how a system reacts to sinusoidal inputs of different frequencies. It provides essential insight into how the amplitude and phase of the output signal vary with frequency, which in turn helps us assess properties such as stability, performance, and robustness.

2 Frequency Response

Back in Exercise 5, we introduced the concept of the transfer function $G(s)$ of a system:

$$G(s) = C(sI - A)^{-1}B + D \quad , s \in \mathbb{C}$$

Remember that we also saw how the steady-state response of a LTI system to a general input $u(t) = e^{st}$ was given by:

$$y_{ss} = G(s)e^{st}$$

In particular, we chose the input e^{st} because almost every input can be expressed as a linear combination of such exponentials.
Then, since we mainly look at LTI systems, we can use the principle of superposition and calculate the response to each exponential separately, then sum them up to get the total response.

Now, let's focus on sinusoidal inputs, which are a specific case of exponentials. A sinusoidal input can be represented as:

$$u(t) = cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

where ω is the angular frequency.

As we said, we can express the input $u(t)$ as a combination of exponentials. Therefore, we get:

$$u(t) = \sum_i U_i e^{s_i t} \qquad \text{with} \quad U_{1,2} = \frac{1}{2}, \quad s_{1,2} = \pm j\omega$$

Using the linearity of the system, we can find the steady-state output as:

$$y_{ss}(t) = \sum_i G(s_i) U_i e^{s_i t} = \frac{1}{2} G(j\omega) e^{j\omega t} + \frac{1}{2} G(-j\omega) e^{-j\omega t}$$

We also saw that we can rewrite $G(j\omega)$ in polar form as:

$$G(j\omega) = |G(j\omega)| e^{j\angle G(j\omega)}$$

where $|G(j\omega)|$ is the magnitude and $\angle G(j\omega)$ is the phase of the transfer function at frequency ω.

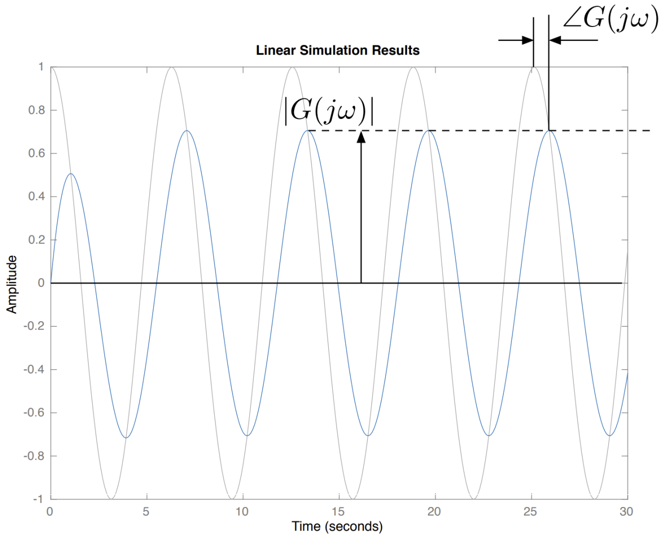
Substituting this into our expression for $y_{ss}(t)$, we get:

$$y_{ss}(t) = \frac{1}{2} |G(j\omega)| e^{j(\omega t + \angle G(j\omega))} + \frac{1}{2} |G(j\omega)| e^{-j(\omega t + \angle G(j\omega))}$$

That we can rewrite as:

$$y_{ss}(t) = |G(j\omega)| cos(\omega t + \angle G(j\omega))$$

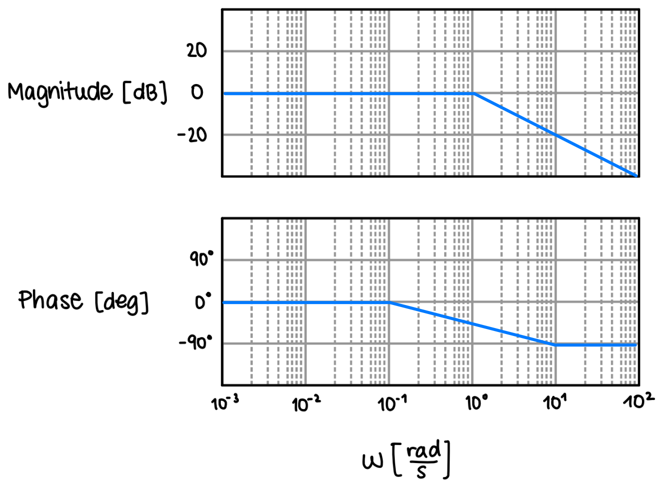
This shows that the output is also a sinusoid at the same frequency ω, but with an amplitude scaled by $|G(j\omega)|$ and a phase shift of $\angle G(j\omega)$.



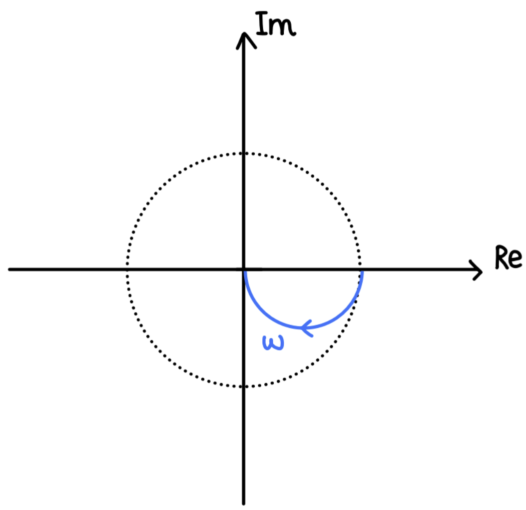
This relationship between the input and output sinusoids is fundamental in understanding the frequency response of LTI systems.
By only analyzing the the magnitude $M = |G(j\omega)|$ and phase shift $\phi = \angle G(j\omega)$ of the transfer function $G(j\omega)$, we can completely characterize how the system responds to sinusoidal inputs at different frequencies.
This is also called the **frequency response** of the system. The frequency response is crucial in control systems engineering, as it helps us to assess system stability, performance, robustness, and many other things of LTI systems.

When representing the frequency response, we are essentially plotting a complex function $G(j\omega) \in \mathbb{C}$ with a single real argument $\omega \in \mathbb{R}$. To visualize this effectively, we have two main graphical methods:

- Bode Plot
  Two separate plots showing the magnitude $|G(j\omega)|$ (in dB) and phase $\angle G(j\omega)$ (in degrees) as functions of frequency ω (on a logarithmic scale).



- Polar Plot (Nyquist Plot)
  Parametric curve showing the evolution of $G(j\omega)$ in the complex plane as ω varies.



2.1 Bode Plot

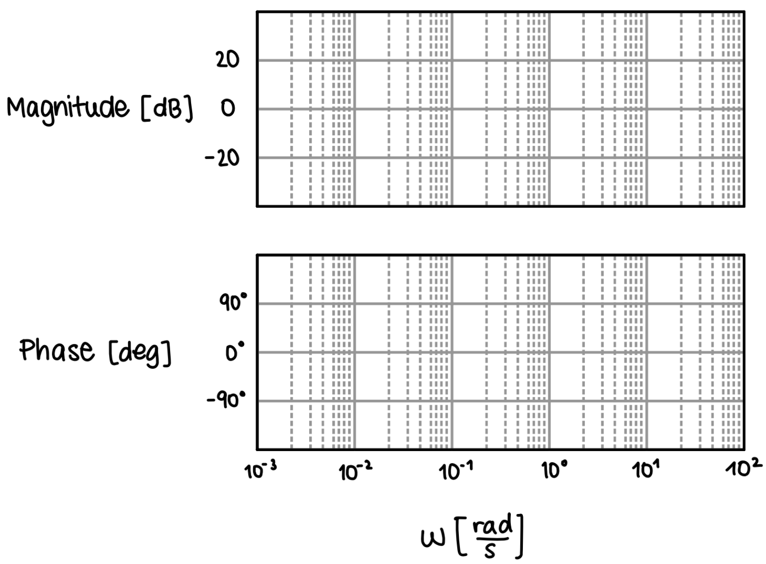
As we said, the Bode Plot consists of **two separate plots**: one for the **magnitude** and one for the **phase** of the transfer function $G(j\omega)$.
The magnitude plot shows how much the system amplifies or attenuates signals at different frequencies, while the phase plot indicates how much the system shifts the phase of the input signal at those frequencies.
Both plots use a logarithmic scale $\log_{10}$ for the frequency axis (in rad/s).

The magnitude is expressed in decibels (dB), calculated as:

$$|G(j\omega)[\text{dB}] = 20\log_{10}|G(j\omega)|$$

x	$\frac{1}{1000}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	2	1	$\sqrt{2}$	10	100
x_{dB}	-60	-40	-20	≈ -6	≈ -3	0	≈ 3	≈ 6	$+20$	$+40$

The phase is expressed in degrees.
All together, the Bode plot has the following representation:



Note that this scaling was chosen because phases just add up when multiplying transfer functions, and in dB (i.e. in log scale) magnitudes also add up when multiplying transfer functions (instead of multplying them as in the linear scale).
Also, inverting the transfer function is equivalent to the reflection of the Bode plot around 0 dB for the magnitude and around 0 degrees for the phase.

To construct the Bode plot for a given transfer function, we typically follow these rules:

Element type	**Formally**	**Magnitude**	**Phase**
single stable pole	$\mathbb{R}(p_i) \leq 0$	$-20\,\frac{\text{dB}}{\text{dec}}$	$-90°$
single unstable pole	$\mathbb{R}(p_i) > 0$	$-20\,\frac{\text{dB}}{\text{dec}}$	$+90°$
complex stable pole	$\mathbb{R}(p_i) \leq 0$	$-40\,\frac{\text{dB}}{\text{dec}}$	$-180°$
complex unstable pole	$\mathbb{R}(p_i) > 0$	$-40\,\frac{\text{dB}}{\text{dec}}$	$+180°$
single min. phase zero	$\mathbb{R}(z_i) \leq 0$	$+20\,\frac{\text{dB}}{\text{dec}}$	$+90°$
single non-min. phase zero	$\mathbb{R}(z_i) > 0$	$+20\,\frac{\text{dB}}{\text{dec}}$	$-90°$
complex non-min. phase zero	$\mathbb{R}(z_i) > 0$	$+40\,\frac{\text{dB}}{\text{dec}}$	$-180°$
complex min. phase zero	$\mathbb{R}(z_i) \leq 0$	$+40\,\frac{\text{dB}}{\text{dec}}$	$+180°$
positive constant	$k \geq 0$	$0\,\frac{\text{dB}}{\text{dec}}$	$0°$
negative constant	$k < 0$	$0\,\frac{\text{dB}}{\text{dec}}$	$\pm180°$
differentiator	$\tau \cdot s$	$+20\,\frac{\text{dB}}{\text{dec}}$	$+90°$
integrator	$\frac{1}{\tau \cdot s}$	$-20\,\frac{\text{dB}}{\text{dec}}$	$-90°$
time delay	$e^{-s\tau}$	$0\,\frac{\text{dB}}{\text{dec}}$	$-\omega\tau$

This said, the overall Bode plot of a transfer function can be constructed by summing the individual contributions of each element (poles, zeros, constants, etc.) in the system.
Remember that when drawing the Bode plot, we need to have the transfer function written in a specific form that we already discussed, that is the Bode form.

$$G(s) = \frac{k_{bode}}{s^q} \frac{\left(\frac{s}{-z_1}+1\right)\left(\frac{s}{-z_2}+1\right)\dots\left(\frac{s}{-z_m}+1\right)}{\left(\frac{s}{-p_1}+1\right)\left(\frac{s}{-p_2}+1\right)\dots\left(\frac{s}{-p_{n-q}}+1\right)}$$

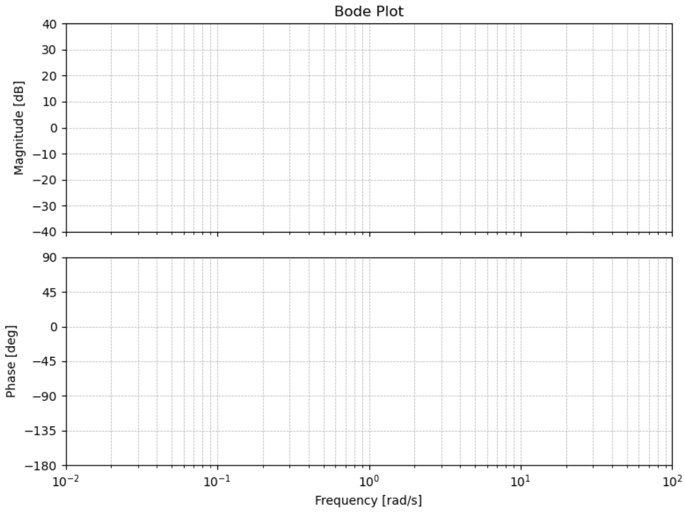
To see how to get into Bode form, rewatch Exercise 6.

Let's see some examples to understand this better.

Example 1

$$G(s) = k \quad \text{with } k = 10$$

Magnitude in dB : $20\log_{10}(10) = 20$ dB



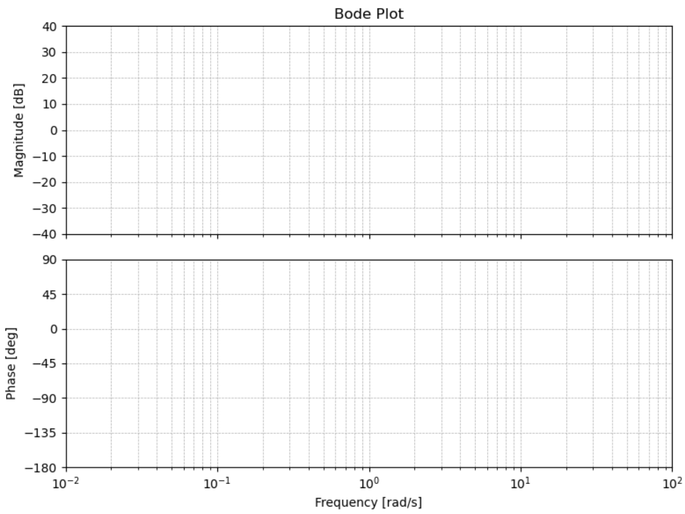
Example 2

$$G(s) = \frac{1}{s+1}$$

Here, to draw the Bode plot, we identify a single stable pole at $s = -1$. Then we take the absolute value of the pole and mark it on the plot.

We draw the low frequency asymptote (before the pole) as a horizontal line at 0 dB. After the pole, we continue with the high frequency asymptote, which has a slope of -20 dB/decade.

For the phase plot, it usually changes 1 decade before and 1 decade after the pole location.

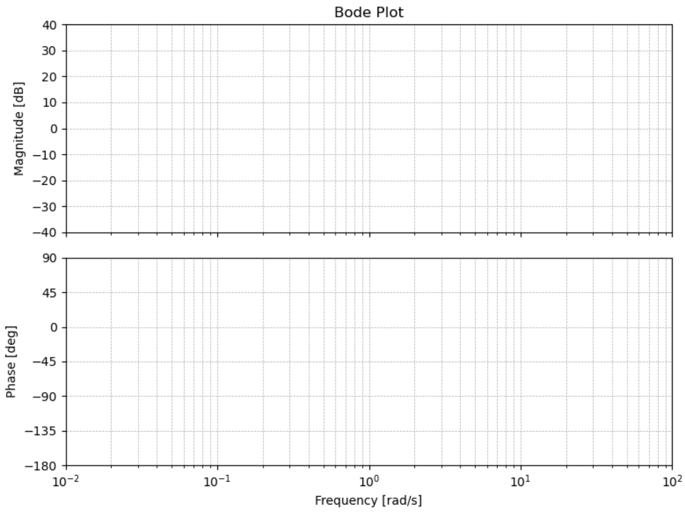


Example 3

Now, as we said earlier, we can combine multiple transfer functions together and draw the overall Bode plot by summing the individual contributions.

$$G(s) = 10\frac{1}{s+1}$$

Since we have the log scale for the magnitude plot, we can simply add the contributions of the constant 10 and the pole at $s = -1$. Same for the phase plot.

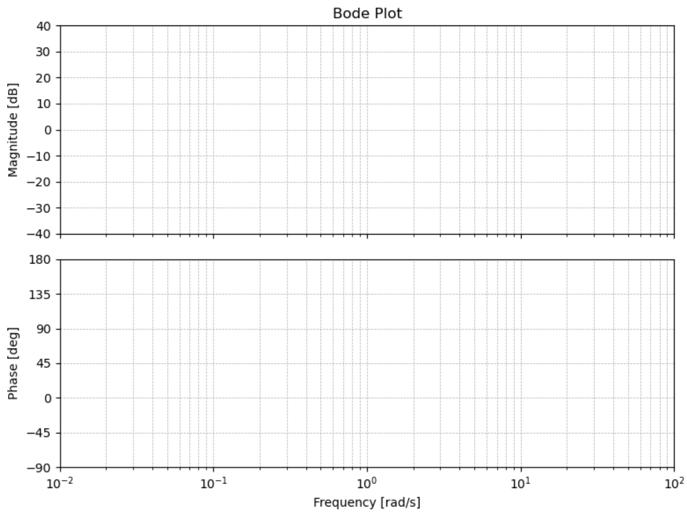


Example 4

We can also have zeros in the transfer function.

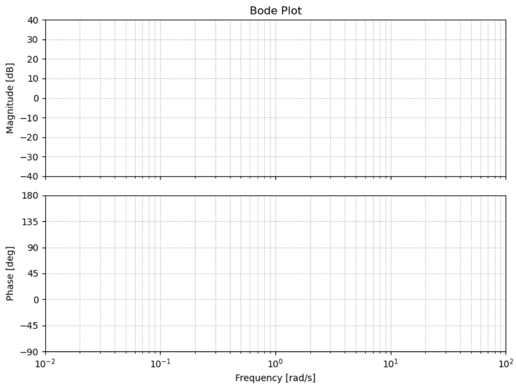
$$G(s) = 10\,\frac{s-10}{s+1} \quad \Longrightarrow \quad G(s) = -100\,\frac{\left(\frac{s}{-10}+1\right)}{s+1}$$

Now we proceed as before, by looking at the contributions of each element from the table.

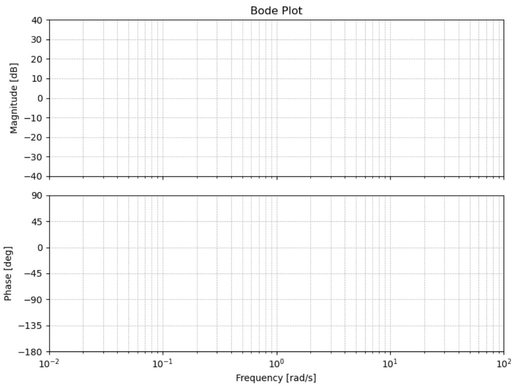


All this can be done for any transfer function, with any number of poles and zeros. Two important cases to consider are the presence of poles and zeros at the origin (i.e., at $s = 0$). These correspond to integrators and differentiators, respectively, and have specific effects on the Bode plot as indicated in the table above.

$$G(s) = s$$

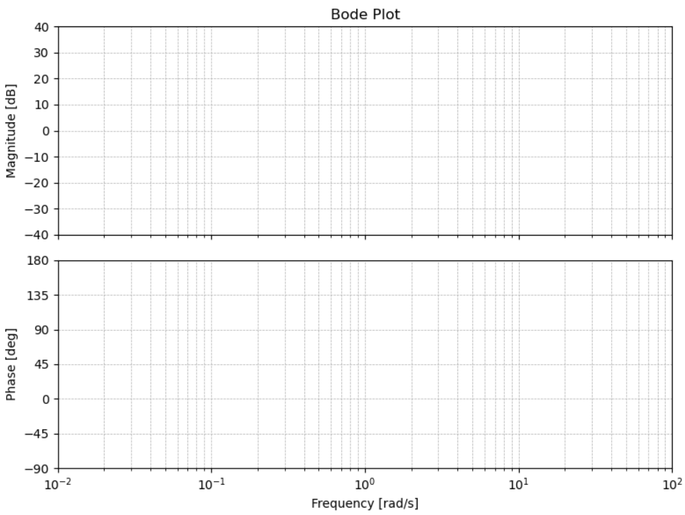


$$G(s) = \frac{1}{s}$$



Example 5

$$G(s) = 20\,\frac{\left(\frac{s}{5}+1\right)\left(\frac{s}{20}+1\right)}{\left(\frac{s}{0.1}+1\right)(s+1)}$$

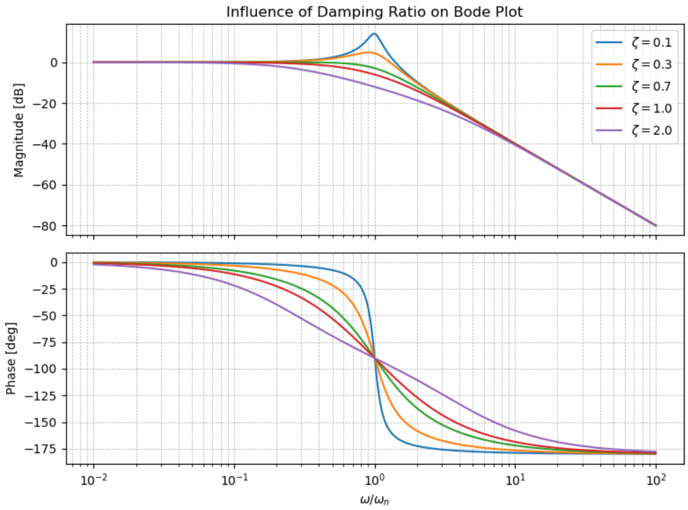


Until now, we have only seen real poles. However, we know that complex poles can also exist. So what happns in that case?
We can generally consider a transfer function of this form:

$$G(s) == \frac{1}{\frac{s^2}{\omega_n^2} + 2\frac{\zeta s}{\omega_n} + 1} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

For these cases, the damping ratio ζ plays a crucial role in determining the shape of the Bode plot.

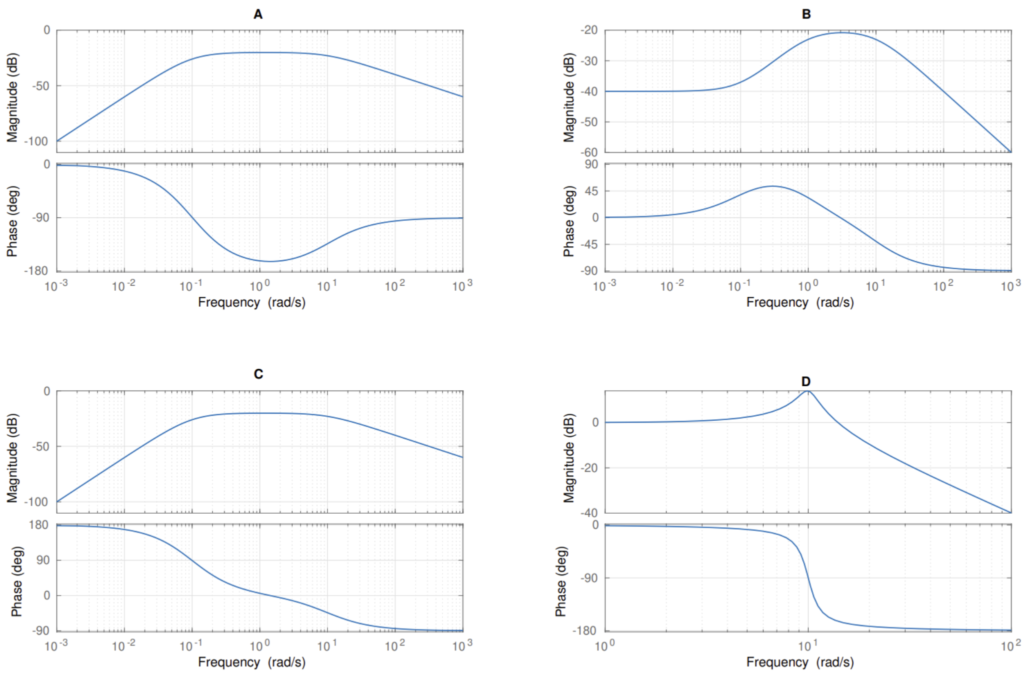
Specifically, for low damping ratios, we observe a resonant peak in the magnitude plot, while for higher damping ratiosm we get an overdamped response without a peak. The phase plot also shows a more rapid transition for lower damping ratios.



Despite this, we can still use the same rules from the table to construct the Bode plot, keeping in mind the influence of the damping ratio on the shape of the plots.

Exam Problem - FS24

Problem: Given in Figure 5 are the Bode plots of four different transfer functions.



Q20 **(1.5 Points)** Assign each transfer function to the corresponding Bode plot.

Transfer Function	A	B	C	D
$L_1(s) = \dfrac{100}{s^2+2s+100}$				
$L_2(s) = \dfrac{s^2}{(s-10)(s+0.1)^2}$				
$L_3(s) = \dfrac{s^2}{(s+10)(s+0.1)^2}$				
$L_4(s) = \dfrac{s+0.1}{(s+1)(s+10)}$				

Exam Problem - HS23

Problem: Consider the transfer functions,

$$G_1(s) = \frac{s+0.1}{s^2},\ G_2(s) = \frac{s+1}{(s-0.1)(s+10)},\ G_3(s) = \frac{1}{s^2+0.02s+1},\ G_4(s) = \frac{s+1}{(s+0.1)(s+10)}.$$

In Figure 13 the Bode plots of the aforementioned transfer functions are shown in a randomized order.

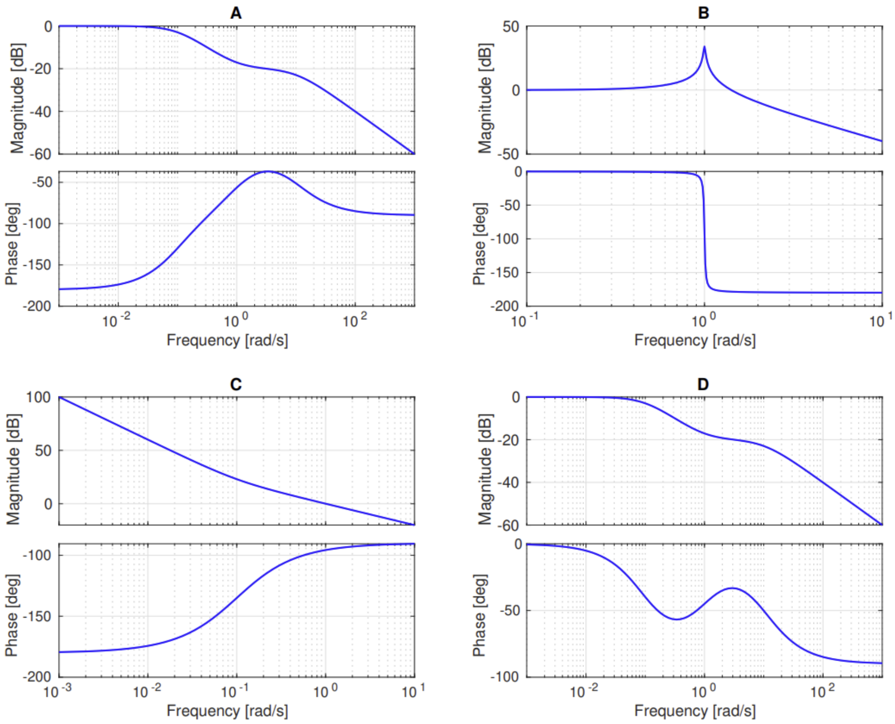


Figure 13: Bode plots of G_1, G_2, G_3 and G_4 in random order.

Q27 **(1 Points)** Mark the correct answer.
Mark the correct Bode plot and transfer function pairings?

- A (A, G_1), (B, G_3), (C, G_2), (D, G_4)
- B (A, G_4), (B, G_3), (C, G_1), (D, G_2)
- C (A, G_2), (B, G_3), (C, G_1), (D, G_4)
- D (A, G_3), (B, G_2), (C, G_4), (D, G_1)

2.2 Polar Plot

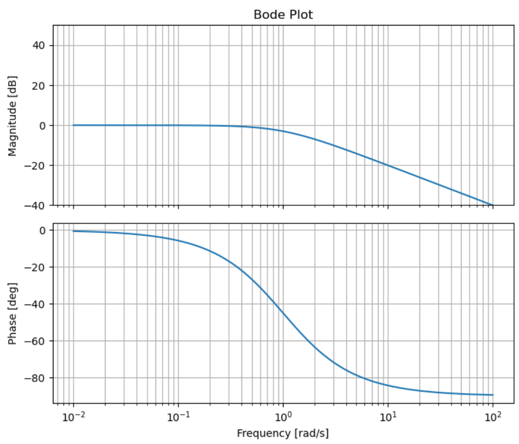
The polar plot, that will bring us to the Nyquist plot next week, is another graphical representation of the frequency response of a system. Unlike the Bode plot, which separates magnitude and phase into two plots, the polar plot combines both into a single plot in the complex plane.
In this representation, we plot the complex values of $G(j\omega)$ as the frequency ω varies from 0 to ∞.
Let's understand it better with at a few examples.

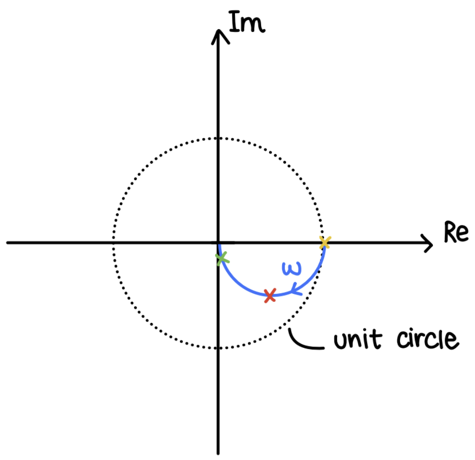
Example 1

We consider the transfer function

$$G(s) = \frac{1}{s+1}$$

To construct the polar plot, we evaluate $G(j\omega)$ for various values of ω by looking at its Bode plot.





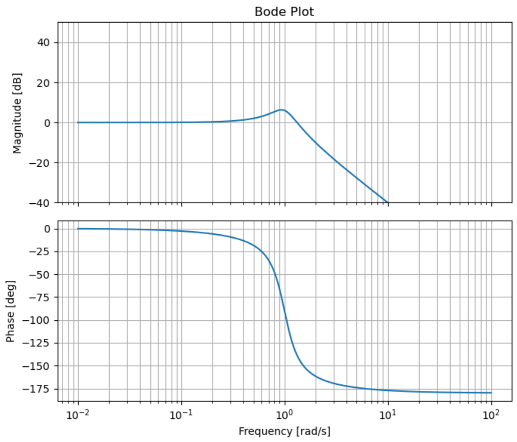
Let's analyze the behavior at key frequencies:

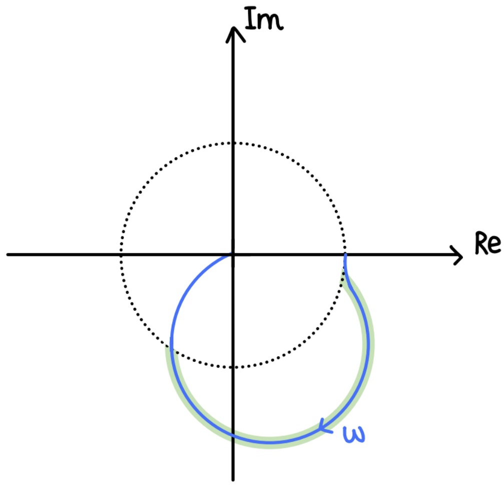
- At low frequencies ($\omega \to 0$): The magnitude is close to 0 dB (or 1 in linear scale), and the phase is close to 0°. Thus, the point on the polar plot is near $(1, 0)$ in the complex plane.
- At the corner frequency ($\omega = 1$ rad/s): The magnitude drops to approximately −3 dB (or about 0.707 in linear scale), and the phase is around −45°. This corresponds to a point on the polar plot at approximately $(0.5, -0.5)$.
- At high frequencies ($\omega \to \infty$): The magnitude approaches −20 dB/decade (or 0 in linear scale), and the phase approaches −90°. Thus, the point on the polar plot approaches the origin along the negative imaginary axis.

Example 2

Now, let's consider a more complex transfer function:

$$G(s) = \frac{1}{s^2 + 0.5s + 1}$$





Example 3

Finally, let's analyze a transfer function with an integrator:

$$G(s) = \frac{1}{s(s^2 + 0.5s + 1)}$$